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Instructions



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How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

havent started

not bad fine no

good thank u how about u

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Today's agenda

- Total Recall
- Spring Framework
- Quiz
- Questions

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What do you remember from the previous workshop ?

http

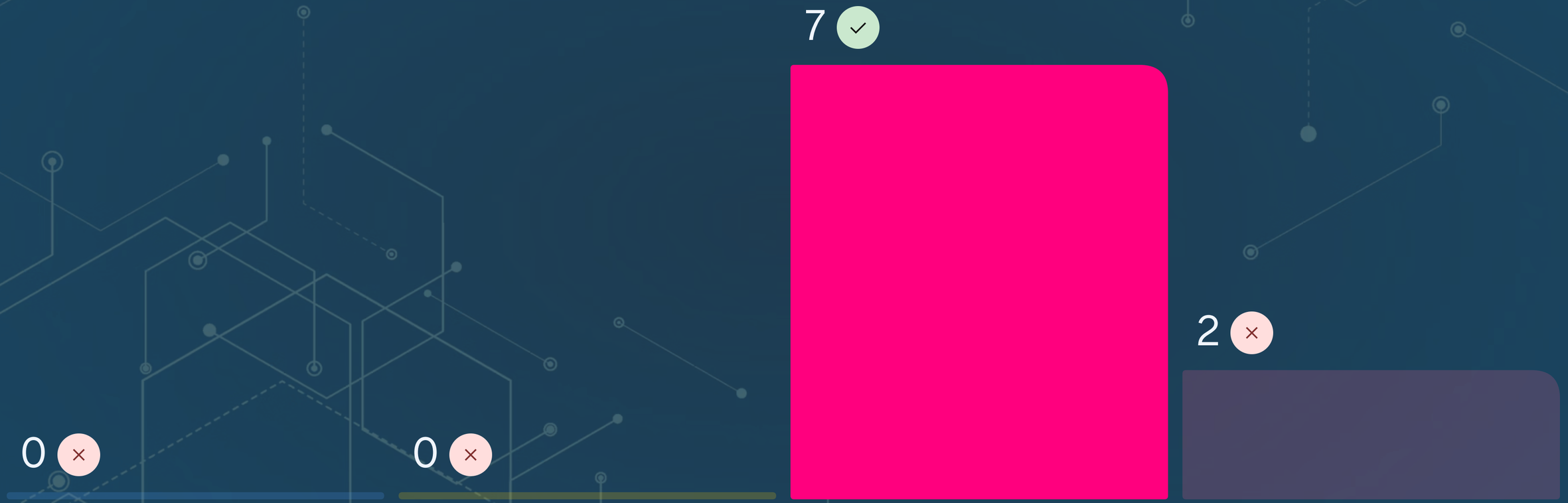
servlets filters

JPA, Servlets

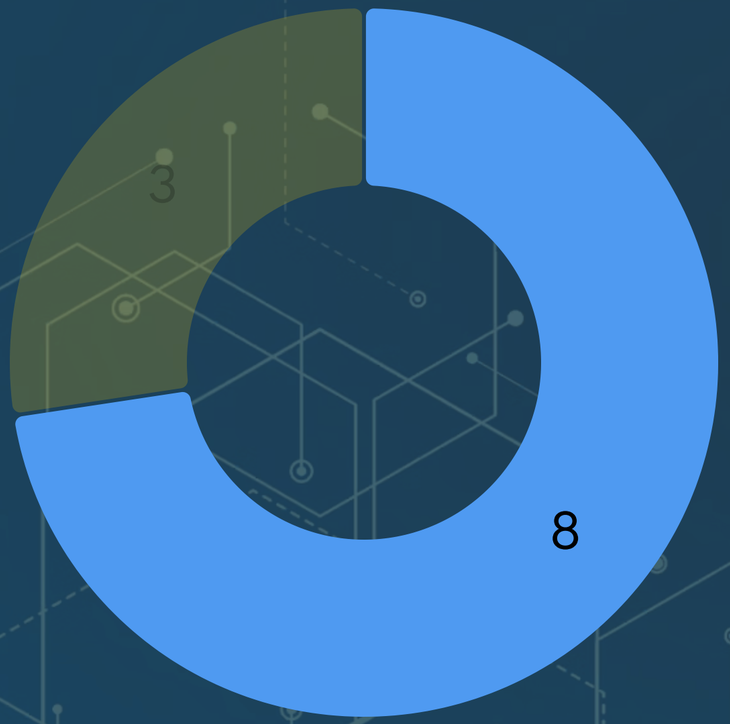
midterm preparation, servlet, tomcat

much time passed, i hardly remember anything

Which interface must be implemented by a filter in Java servlets?

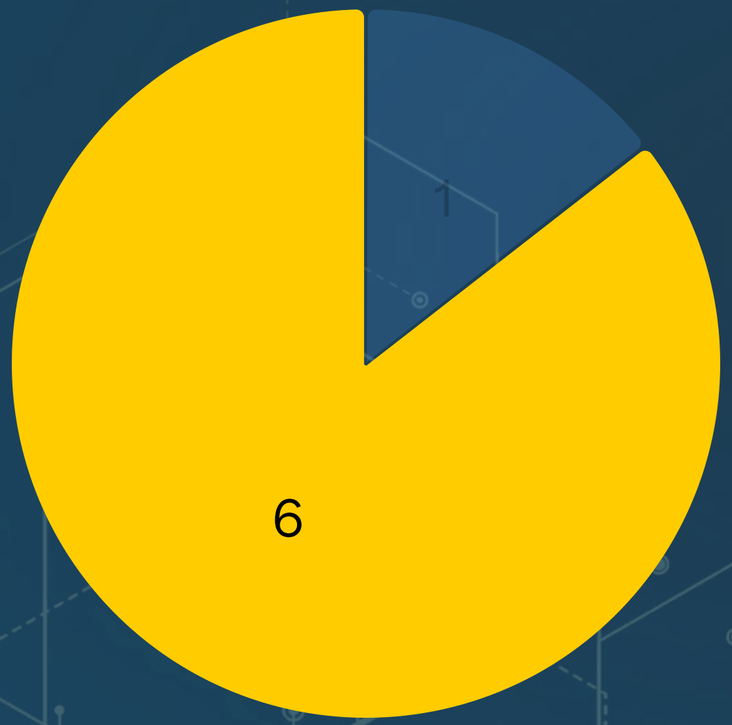


How do you map a filter to a specific servlet or URL pattern?



8	A) By using `<filter-mapping>` in web.xml.	✓
3	B) By adding an annotation in the servlet code.	×
0	C) By creating a database entry for the filter.	×
0	D) By configuring the filter in the session object.	×

Where do you configure listeners in a Java web application?



1	A) In the servlet code.	✗
6	B) In the web.xml deployment descriptor file.	✓
0	C) In the HTTP header configuration.	✗
0	D) In the application's root directory.	✗

What does JSTL stand for in the context of JSP?



8 A) Java Standard Tag Library. ✓

1 B) Java Server Template Library. ×

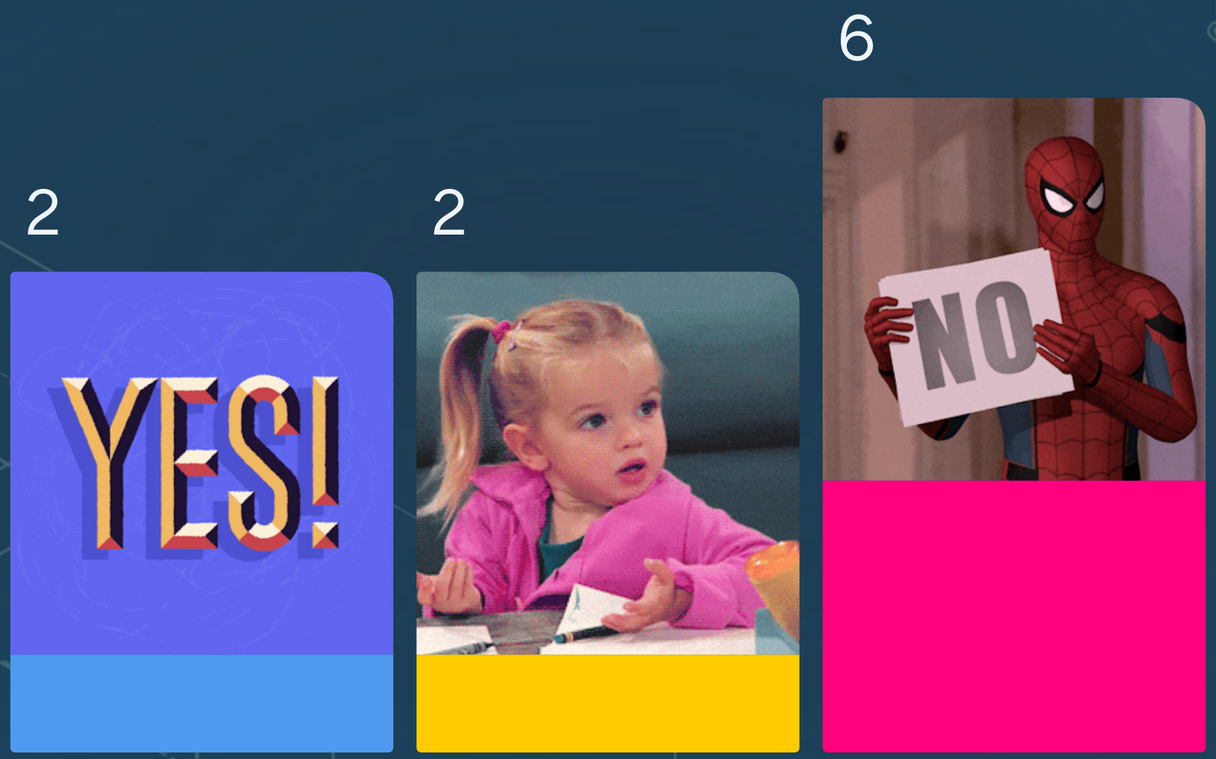
0 C) JSP Standard Template Library. ×

1 D) JSP Servlet Tag Library. ×

Spring Core (ws-15)

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Have you started 'Spring Core'?



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IoC, DIP, DI

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👑 Inversion of Control (IoC) — *The Philosophy*

- **What it means:** "Don't call us, we'll call you" (The Hollywood Principle).
- **Role:** This is the broadest conceptual umbrella. It simply means taking control of program flow, lifecycle management, or object creation away from your custom application code and moving it to a framework, container, or system driver.
- *Examples of IoC*: Event loop handling in a browser, UI framework lifecycle hooks, and object management.

📐 Dependency Inversion Principle (DIP) — *The Strategy*

- **What it means:** "Depend on abstractions, not concretions."
- **Role:** A sub-concept under IoC focused strictly on **structural design and decoupling code modules**. It is a rule that guides how you write your code so that high-level business logic is insulated from low-level infrastructure details.

🛠️ Dependency Injection (DI) — *The Tactic / Implementation Tool*

- **What it means:** "Pass dependencies in from the outside."
- **Role:** A highly specific creational pattern used to physically supply an object with its dependencies. It is the most common way to fulfill the structural requirements of DIP and realize the lifecycle management of an IoC container.

📦 IoC / DI Container — *The Infrastructure*

- **What it means:** "The automated warehouse that builds and delivers the objects."
- **Role:** The actual framework runtime component (like the Spring ApplicationContext, .NET ServiceProvider, or Guice Injector) that automates the instantiation and wiring of your classes so you don't have to manually call `new` or stitch constructors together.

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Spring

practice...

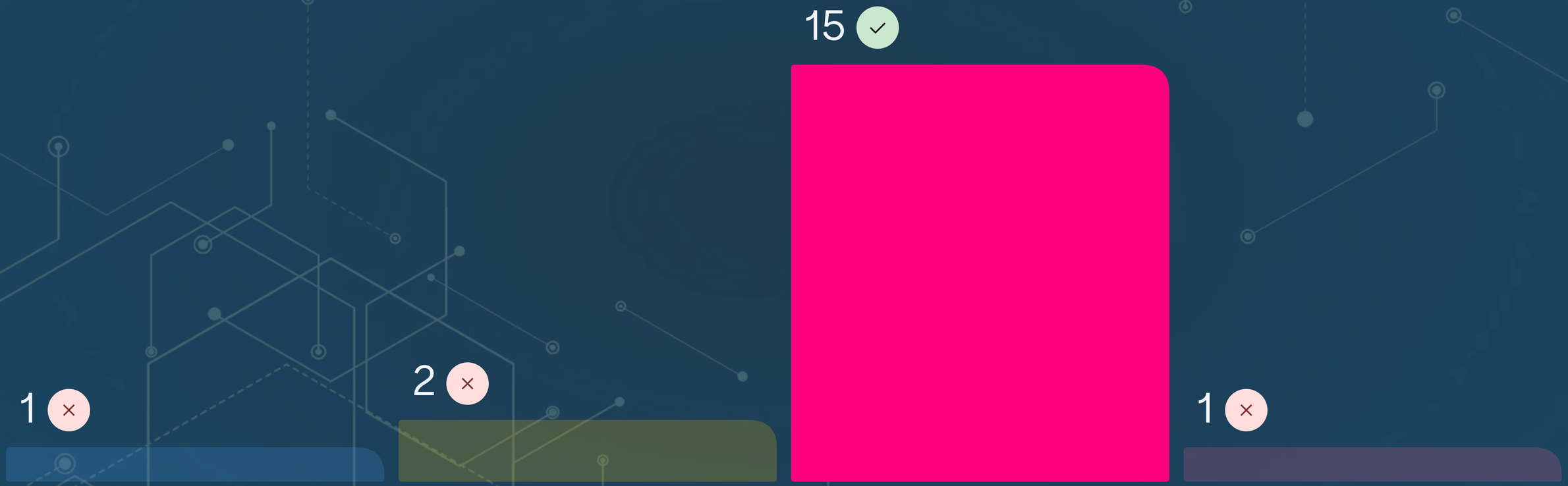
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Quiz

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What is core component of Spring Framework that provides the basic functionality for configuration and dependency injection?



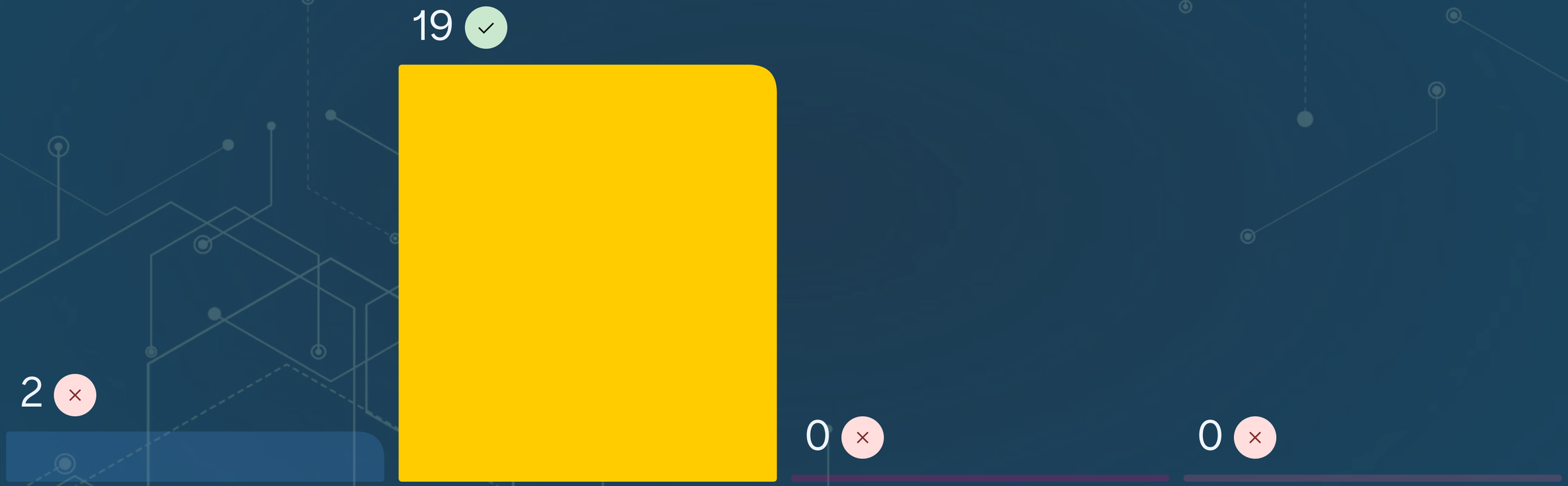
A) Hibernate.

B) Spring MVC.

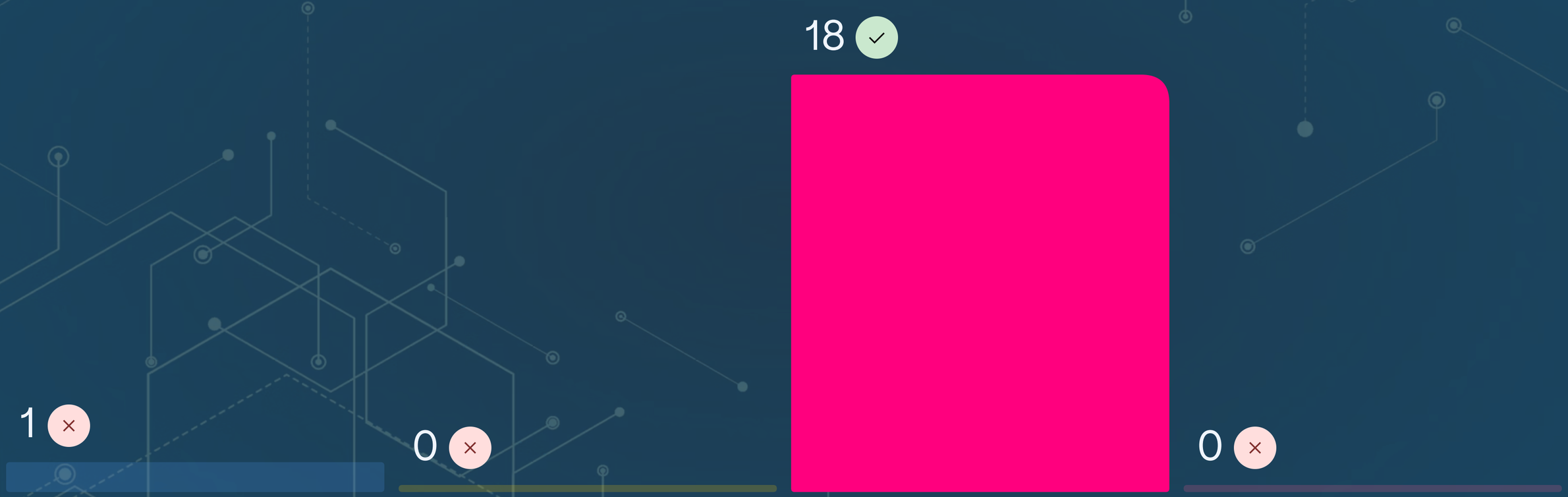
C) Spring Container.

D) Spring AOP.

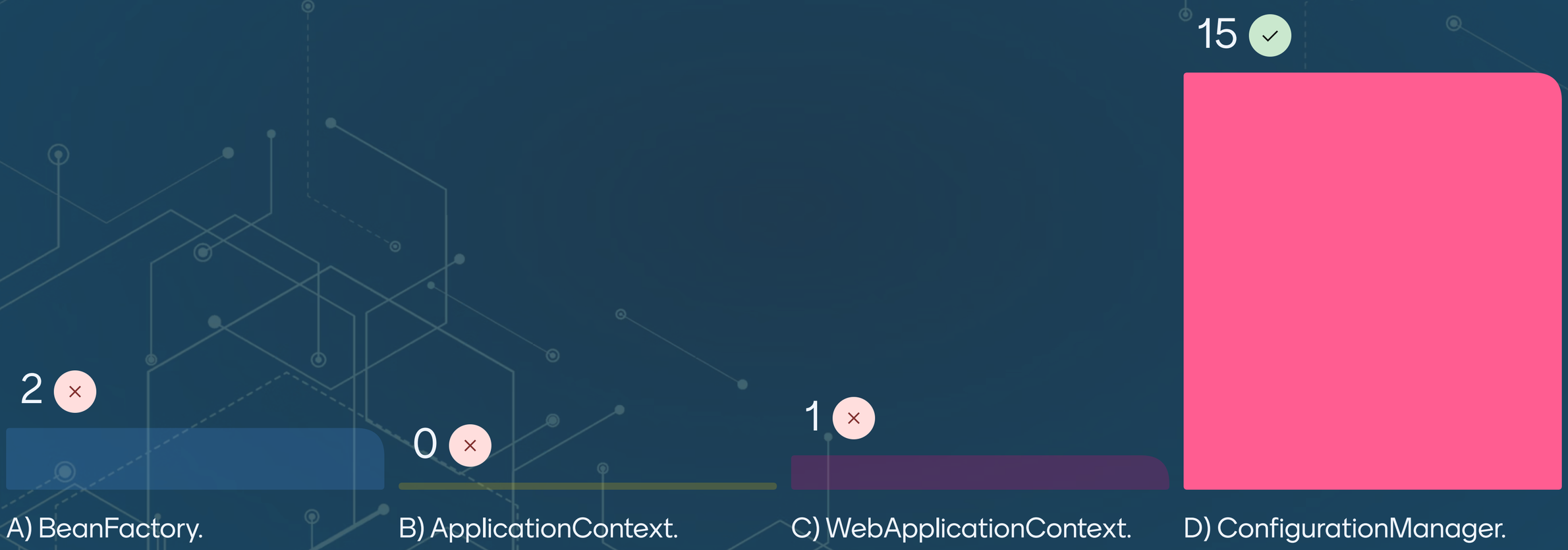
Which concept in Spring Architecture is primarily responsible for simplifying the development of loosely coupled applications?



What does the term "IoC" stand for in the Spring IoC Container?



Which of the following is NOT a type of IoC container provided by the Spring Framework?



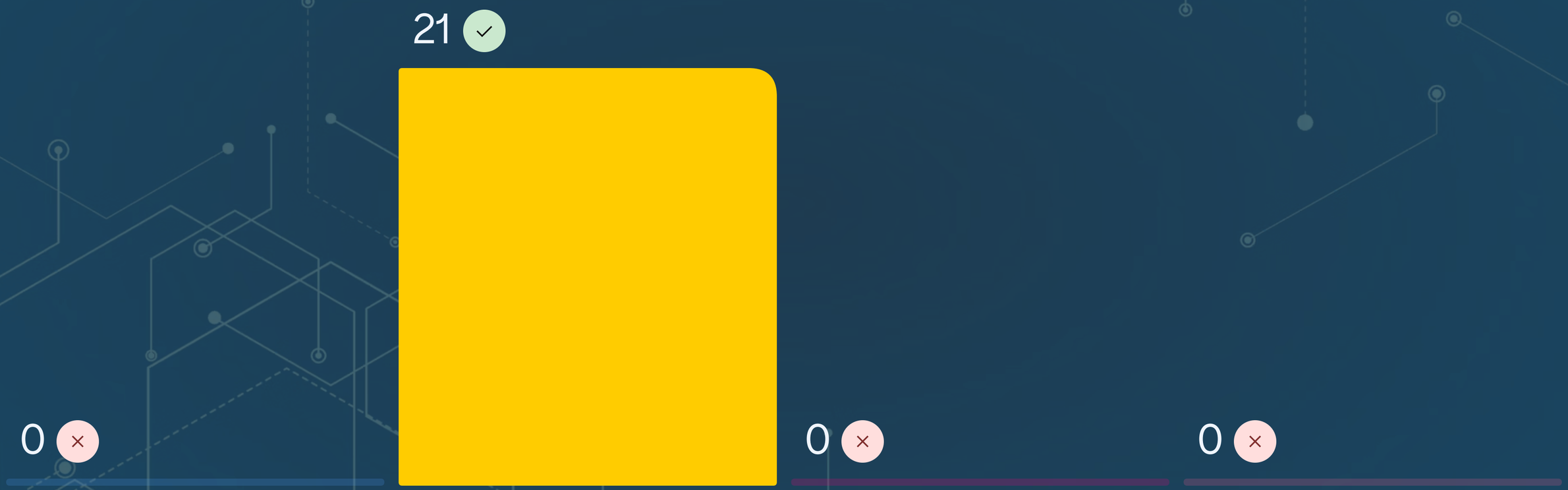
A) BeanFactory.

B) ApplicationContext.

C) WebApplicationContext.

D) ConfigurationManager.

What is the primary role of the Spring IoC container?



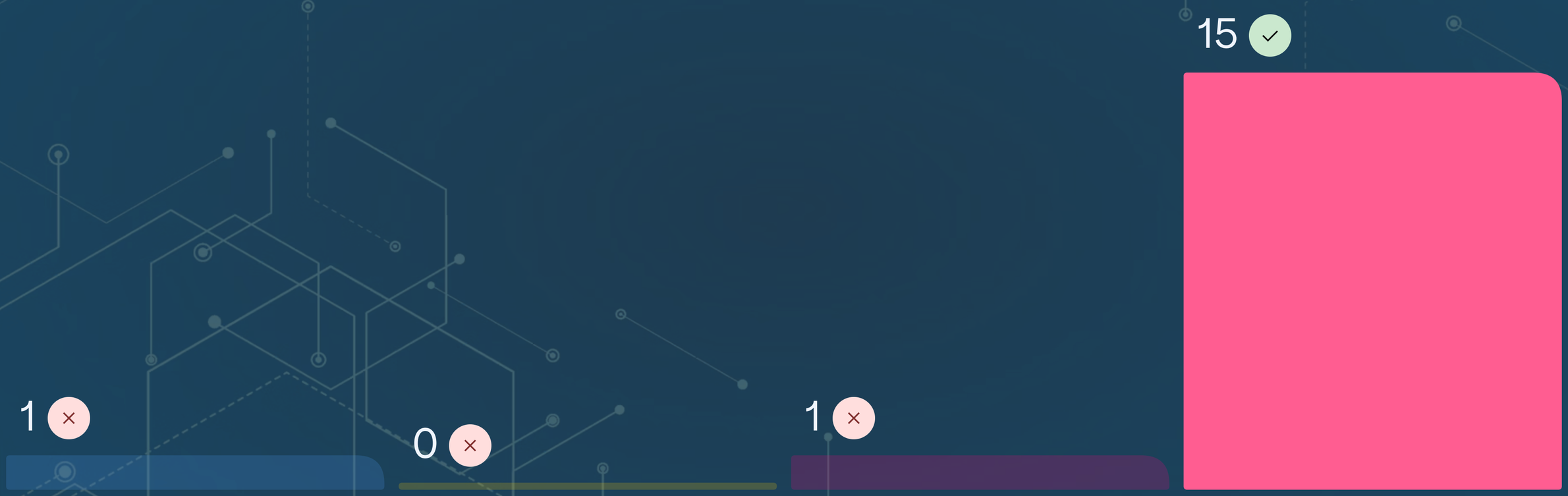
A) To manage cross-cutting concerns like logging and transactions.

B) To instantiate, configure, and manage the lifecycle of application

C) To improve application performance with caching.

D) To handle REST API requests.

Which configuration approach can be used to define beans for the IoC container in Spring?



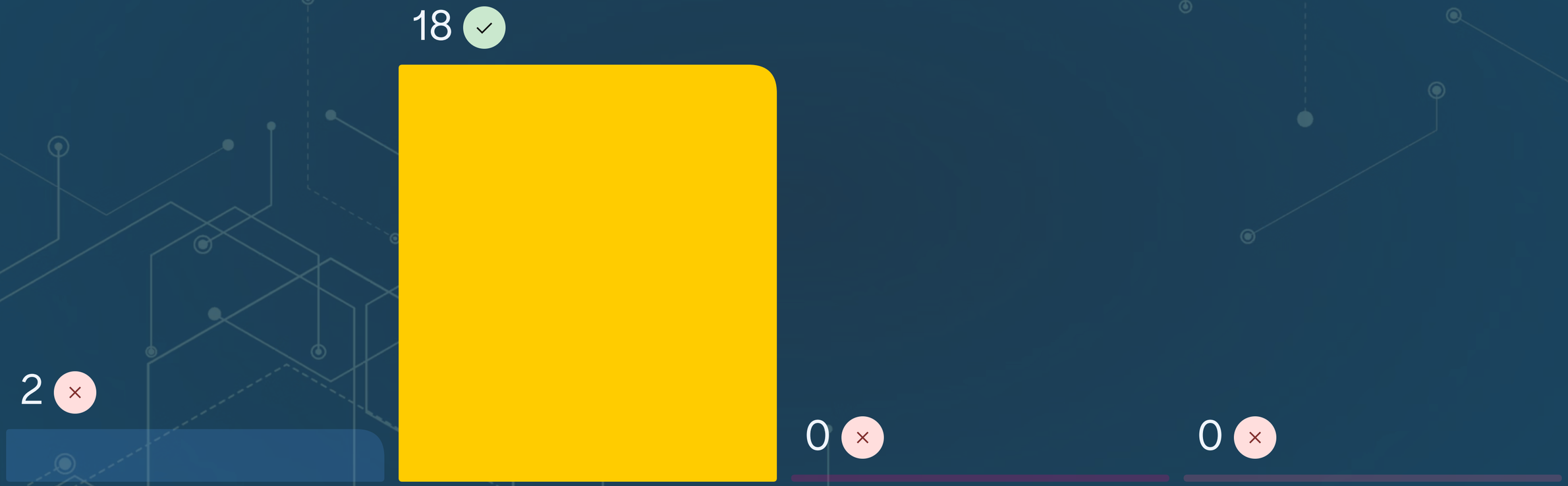
A) XML-based configuration.

B) Annotation-based configuration.

C) Java-based configuration.

D) All of the above.

What is Dependency Injection in the context of the Spring Framework?



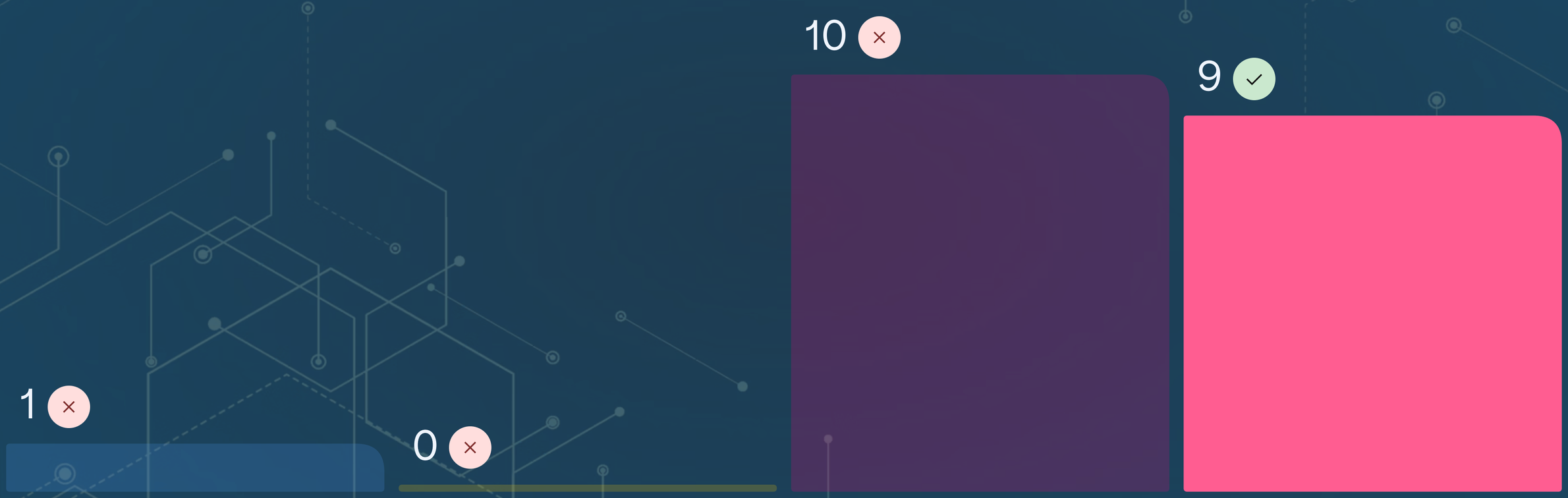
A) A technique to handle database transactions.

B) A design pattern used to inject dependencies into an object.

C) A testing framework for Spring-based applications.

D) A Spring module for creating web applications.

Which of the following is NOT a method of implementing dependency injection in Spring?



A) Setter Injection.

B) Constructor Injection.

C) Method Injection.

D) Property Injection (Java Reflection).

What is the main benefit of using Dependency Injection in Spring?



20 ✓

A) Achieving loose coupling between components.

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B) Faster execution of the application.

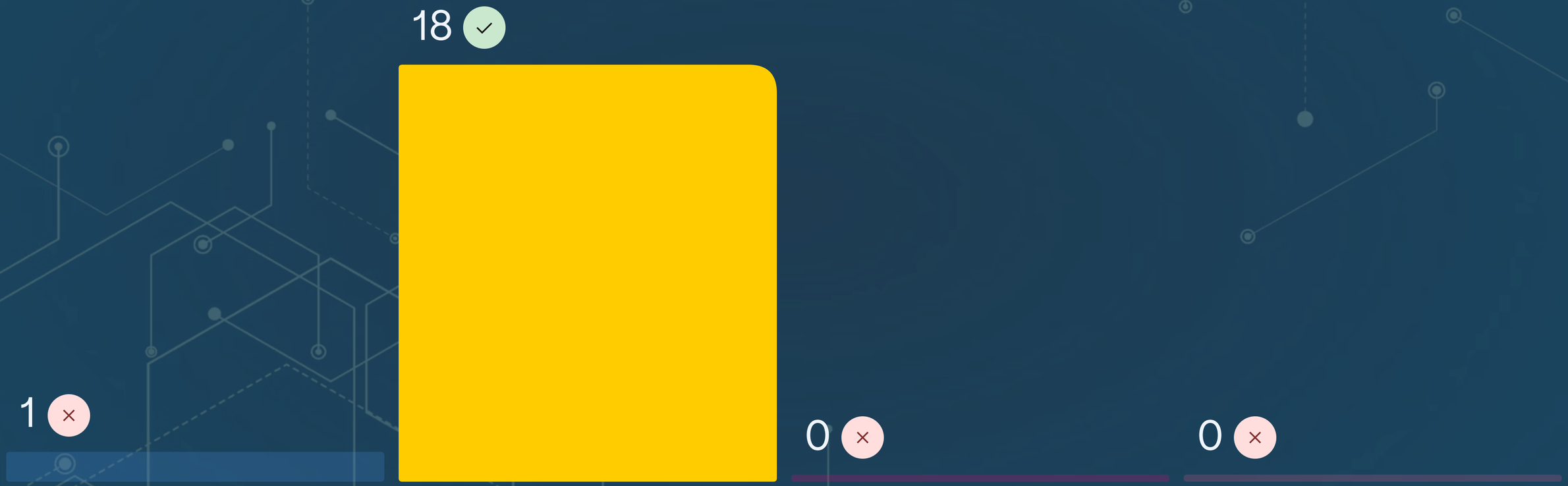
0 ×

C) Simplifying database interactions.

0 ×

D) Automatically generating REST APIs.

In annotation-based configuration, which annotation marks a class as a Spring-managed bean that can have its dependencies injected?



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Quiz leaderboard



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Next time: Spring Core

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Questions?

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